

Investering og finansiering aflevering 1

We assume that the Markowitz framework holds

$$\bar{r} = \begin{pmatrix} 0.3 \\ 0.2 \\ 0.15 \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} 0.25 & 0.12 & 0.04 \\ 0.12 & 0.16 & 0.05 \\ 0.04 & 0.05 & 0.08 \end{pmatrix}$$

We do some prep work by calculating the different factors

$$a = (1 \ 1 \ 1)\Sigma^{-1}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = (1 \ 1 \ 1)\begin{pmatrix} 0.25 & 0.12 & 0.04 \\ 0.12 & 0.16 & 0.05 \\ 0.04 & 0.05 & 0.08 \end{pmatrix}^{-1}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 13,9$$

$$b = (1 \ 1 \ 1)\Sigma^{-1}\bar{r} = (1 \ 1 \ 1)\begin{pmatrix} 0.25 & 0.12 & 0.04 \\ 0.12 & 0.16 & 0.05 \\ 0.04 & 0.05 & 0.08 \end{pmatrix}^{-1}\begin{pmatrix} 0.3 \\ 0.2 \\ 0.15 \end{pmatrix} = 2,39$$

$$c = \bar{r}^T \Sigma^{-1} \bar{r} = (0.3 \ 0.2 \ 0.15)\begin{pmatrix} 0.25 & 0.12 & 0.04 \\ 0.12 & 0.16 & 0.05 \\ 0.04 & 0.05 & 0.08 \end{pmatrix}^{-1}\begin{pmatrix} 0.3 \\ 0.2 \\ 0.15 \end{pmatrix} = 0,503$$

$$d = a \cdot c - b^2 = 13,9 \cdot 0,503 - 2,39^2 \approx 1,28$$

- 1) Calculate the weights, variances and return of the minimum variance portfolio,

We have formulas which makes it easy

$$V_{mvp} = \frac{1}{a} = \frac{1}{13,9} \approx 0,0719$$

$$w_{mvp} = \frac{1}{a} \cdot \Sigma^{-1} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 0,0719 \cdot \begin{pmatrix} 0.25 & 0.12 & 0.04 \\ 0.12 & 0.16 & 0.05 \\ 0.04 & 0.05 & 0.08 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \approx \begin{pmatrix} 0,100 \\ 0,135 \\ 0,764 \end{pmatrix}$$

$$r_{mvp} = \bar{r}^T w_{mvp} = (0.3 \ 0.2 \ 0.15) \begin{pmatrix} 0,100 \\ 0,135 \\ 0,764 \end{pmatrix} = 0,1716$$

A risk free asset enters the market $r_f = 0.03$

- 2) Find the weights of the tangency portfolio

We first find the effective return

$$\bar{r}^e = \bar{r} - \begin{pmatrix} r_f \\ r_f \\ r_f \end{pmatrix} = \begin{pmatrix} 0.3 \\ 0.2 \\ 0.15 \end{pmatrix} - \begin{pmatrix} 0.03 \\ 0.03 \\ 0.03 \end{pmatrix} = \begin{pmatrix} 0,27 \\ 0,17 \\ 0,12 \end{pmatrix}$$

Again we just use a formula to calculate

$$w_{tan} = \frac{\Sigma^{-1} \bar{r}^e}{(1 \ 1 \ 1)\Sigma^{-1} \bar{r}^e} = \frac{\begin{pmatrix} 0.25 & 0.12 & 0.04 \\ 0.12 & 0.16 & 0.05 \\ 0.04 & 0.05 & 0.08 \end{pmatrix}^{-1} \begin{pmatrix} 0,27 \\ 0,17 \\ 0,12 \end{pmatrix}}{(1 \ 1 \ 1)\begin{pmatrix} 0.25 & 0.12 & 0.04 \\ 0.12 & 0.16 & 0.05 \\ 0.04 & 0.05 & 0.08 \end{pmatrix}^{-1} \begin{pmatrix} 0,27 \\ 0,17 \\ 0,12 \end{pmatrix}} = \begin{pmatrix} 0,444 \\ 0,0468 \\ 0,510 \end{pmatrix}$$

- 3) Calculate return and variances of the tangency portfolio

$$\bar{r}_{tan} = \bar{r}^T \cdot w_{tan} = (0.3 \quad 0.2 \quad 0.15) \begin{pmatrix} 0,444 \\ 0,0468 \\ 0,510 \end{pmatrix} \approx 0,219$$

$$V_{tan} = w_{tan}^T \cdot \Sigma \cdot w_{tan} = transpose \left(\begin{pmatrix} 0,444 \\ 0,0468 \\ 0,510 \end{pmatrix} \right) \begin{pmatrix} 0.25 & 0.12 & 0.04 \\ 0.12 & 0.16 & 0.05 \\ 0.04 & 0.05 & 0.08 \end{pmatrix} \cdot \begin{pmatrix} 0,444 \\ 0,0468 \\ 0,510 \end{pmatrix} \approx 0,0959$$

Might as well calculate sd for later

$$\sigma_{tan} = \sqrt{0,0959} \approx 0,310$$

- 4) Write down an equation for the efficient frontier

The efficient frontier after the introduction of the risk free asset is just the line that starts in the risk free asset and hits the tangency portfolio

The line which hits

(0,0.03) and (0.31,0.22)

$$\bar{r}_p(\sigma) = \frac{0.19}{0.31} \cdot \sigma + 0.03 = 0,613 \cdot \sigma + 0,03$$

ASSUME ACPM HOLDS

- 5) What are the weights of the market portfolio

Given CAPM the market portfolio is exactly the tangency portfolio

$$w_M = w_{tan} = \begin{pmatrix} 0.44 \\ 0.05 \\ 0.51 \end{pmatrix}$$

- 6) Calculate the betas of the 3 assets

We calculate our beta vector

$$\beta = \frac{1}{v_m} \Sigma \cdot w_M = \frac{1}{0.096} \begin{pmatrix} 0.25 & 0.12 & 0.04 \\ 0.12 & 0.16 & 0.05 \\ 0.04 & 0.05 & 0.08 \end{pmatrix} \begin{pmatrix} 0.44 \\ 0.05 \\ 0.51 \end{pmatrix} = \begin{pmatrix} 1,42 \\ 0,899 \\ 0,634 \end{pmatrix}$$

- 7) Decompose into systematic and unsystematic risk

Again we have an expression of the risk from slides

$$var(r_i) = \sigma_i^2 = \beta_i^2 \cdot \sigma_m^2 + var(\epsilon_i)$$

σ_i^2	$\beta_i^2 \cdot \sigma_m^2$	$var(\epsilon_i)$
Total risik	Ussystematic risik	"usystematisk risik"
0.25	$1.42^2 \cdot 0.096 \approx 0,194$	$= 0.25 - 0,194 \approx 0,056$
0.16	$0.899^2 \cdot 0.096 \approx 0,0776$	$0.16 - 0,0776 = 0,0824$
0.08	$0.634^2 \cdot 0.096 \approx 0,0386$	$0.08 - 0,0386 = 0,0414$

Opgave 2

3 obligationer er givet med følgende cashflows

$$B = \begin{pmatrix} 105 & 0 & 0 & 0 \\ 3 & 103 & 0 & 0 \\ 0 & 0 & 0 & 100 \end{pmatrix}, P = \begin{pmatrix} 100.96 \\ 98.11 \\ 85.48 \end{pmatrix}$$

- 1) calculate spotrates s_1, s_2 og s_4 based på annual compounding
first we find s_1

$$100.9615 = \frac{105}{1 + s_1}$$

⇕

$$s_1 = 0,04$$

then s_2

$$98.1139 = \frac{3}{1 + 0,0400004} + \frac{103}{(1 + s_2)^2}$$

⇕

$$s_2 = 0,04$$

and s_4

$$85.4804 = \frac{100}{(1 + s_4)^4}$$

$$\Leftrightarrow s_4 = 0,04$$

Then we have

$$s_1 = s_2 = s_4 = 4\%$$

- 2) Find modified duration of the 3 bonds

First we find the yield to maturity for all 3 bonds

Since we exactly did this calculation before

$$\lambda_1 = 4\%$$

For the 2. we use the general equation

$$98.1139 = \frac{\lambda_2: 3}{1 + x} + \frac{103}{(1 + x)^2}$$

⇕

$$x = 0,04$$

$$\lambda_2 = 4\%$$

And since $\lambda = s_t$ for 0 coupon bonds

$$\lambda_4 = 4\%$$

We have constant YTM

Formula for modified duration

$$D_{mod} = \frac{1}{p} \cdot \frac{1}{1 + \lambda} \cdot \sum_{k=1}^n k \cdot \frac{c_k}{(1 + \lambda)^k}$$

We construct a vector with the weighted discounted payments before normalizing with respect to individual prices

$$D_{mod} = \frac{1}{1 + 0.04} \begin{pmatrix} 105 & 0 & 0 & 0 \\ 3 & 103 & 0 & 0 \\ 0 & 0 & 0 & 100 \end{pmatrix} \begin{pmatrix} \frac{1}{1 + 0.04} \\ \frac{2}{(1 + 0.04)^2} \\ \frac{3}{(1 + 0.04)^3} \\ \frac{4}{(1 + 0.04)^4} \end{pmatrix} / P \approx \begin{pmatrix} 97,08 \\ 185,9 \\ 328,8 \end{pmatrix} / P$$

Then we divided

$$D_{mod} = \begin{pmatrix} \frac{97,08}{100,96} \\ \frac{185,9}{185,9} \\ \frac{98,11}{328,8} \\ \frac{328,8}{85,48} \end{pmatrix} \approx \begin{pmatrix} 0,9615689 \\ 1,894812 \\ 3,846514 \end{pmatrix}$$

3) Calculate convexity

Formula:

$$C = \frac{1}{P \cdot (1 + \lambda)^2} \cdot \sum_{k=1}^T k \cdot (k + 1) \cdot \frac{c_k}{(1 + \lambda)^k}$$

We use the exact same method as before

$$C = \left(\frac{1}{1 + 0.04} \right)^2 \begin{pmatrix} 105 & 0 & 0 & 0 \\ 3 & 103 & 0 & 0 \\ 0 & 0 & 0 & 100 \end{pmatrix} \begin{pmatrix} \frac{2}{1 + 0.04} \\ \frac{6}{(1 + 0.04)^2} \\ \frac{12}{(1 + 0.04)^3} \\ \frac{20}{(1 + 0.04)^4} \end{pmatrix} / P \approx \begin{pmatrix} 186,7 \\ 533,6 \\ 1580,6 \end{pmatrix} / P$$

$$C = \begin{pmatrix} \frac{186,7}{100,96} \\ \frac{533,6}{185,9} \\ \frac{98,11}{1580,6} \\ \frac{1580,6}{85,48} \end{pmatrix} \approx \begin{pmatrix} 1,849 \\ 5,439 \\ 18,49 \end{pmatrix}$$

A new bond enters the market with cashflows stated under and $P_4 = 100$

$$B_4 = (4 \quad 4 \quad 104 \quad 0)$$

- 4) Can you use the 3 bonds in the market to hedge a short position in bond 4 against a shift in yield matching both price duration and convexity? Yes since the 3 bonds in the market have linearly independent cash flows we can hedge with respect to price duration and convexity. We will show it by doing it

First we see that face value is the same as price hence

$$\lambda_4 = 4\%$$

Now we calculate the duration and the convexity for the new bond

$$D_4 = \left(\frac{1}{1+0.04} \right)^1 \cdot \frac{1}{100} \begin{pmatrix} 4 & 4 & 104 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{1+0.04} \\ \frac{2}{(1+0.04)^2} \\ \frac{3}{(1+0.04)^3} \\ \frac{4}{(1+0.04)^4} \end{pmatrix} \approx 2,775$$

$$C_4 = \left(\frac{1}{1+0.04} \right)^2 \cdot \frac{1}{100} \begin{pmatrix} 4 & 4 & 104 & 0 \end{pmatrix} \begin{pmatrix} \frac{2}{1+0.04} \\ \frac{6}{(1+0.04)^2} \\ \frac{12}{(1+0.04)^3} \\ \frac{20}{(1+0.04)^4} \end{pmatrix} \approx 10,53$$

We create A matrix that represents the system of equations

$$\begin{aligned} P_4 &= x_1 \cdot P_1 + x_2 \cdot P_2 + x_3 \cdot P_3 \\ P_4 \cdot D_4^m &= x_1 \cdot P_1 \cdot D_1^m + x_2 \cdot P_2 \cdot D_2^m + x_3 \cdot P_3 \cdot D_3^m \\ P_4 \cdot C_4 &= x_1 \cdot P_1 \cdot C_1 + x_2 \cdot P_2 \cdot C_2 + x_3 \cdot P_3 \cdot C_3 \end{aligned}$$

$$\begin{pmatrix} 100,96 & 98,11 & 85,48 \\ 97,07304 & 185,9185 & 328,7561 \\ 186,675 & 533,6203 & 1580,525 \end{pmatrix}$$

Which is invertible thus we can find x_1, x_2, x_3

$$\begin{pmatrix} 100,96 & 98,11 & 85,48 \\ 97,07304 & 185,9185 & 328,7561 \\ 186,675 & 533,6203 & 1580,525 \end{pmatrix}^{-1} \cdot \begin{pmatrix} 100 \\ 100 \cdot 2.775 \\ 100 \cdot 10.53 \end{pmatrix} \approx \begin{pmatrix} -0,2967 \\ 1,011 \\ 0,3599 \end{pmatrix}$$

This gives us our position I

$$I = (-0,2967 \quad 1,011 \quad 0,3599 \quad -1)$$

With the in the market

$$B = \begin{pmatrix} 105 & 0 & 0 & 0 \\ 3 & 103 & 0 & 0 \\ 0 & 0 & 0 & 100 \\ 4 & 4 & 104 & 0 \end{pmatrix}$$

- 5) Let yield increase by 1 percentage point what is the PnL of our position
 Since yield was 4 for all assets before they must all have a yield of $\lambda' = 5\%$
 We find the prices by discounting the cashflows

$$\begin{pmatrix} 105 & 0 & 0 & 0 \\ 3 & 103 & 0 & 0 \\ 0 & 0 & 0 & 100 \\ 4 & 4 & 104 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{1+0.05} \\ \frac{1}{(1+0.05)^2} \\ \frac{1}{(1+0.05)^3} \\ \frac{1}{(1+0.05)^4} \end{pmatrix} \approx \begin{pmatrix} 100 \\ 96,28 \\ 82,27 \\ 97,28 \end{pmatrix}$$

$$Pnl = P_{after} - P_{before}$$

The price before was exactly 0 by the immunization

Thus

$$PnL = (-0,2967 \quad 1,011 \quad 0,3599 \quad -1) \begin{pmatrix} 100 \\ 96,28 \\ 82,27 \\ 97,28 \end{pmatrix} \approx -0,001947$$

Which is roughly 0 which means our immunization did a good job

- 6) An annuity with the following cashflows is introduced and must be priced fairly

$$B_{ann} = (50 \quad 50 \quad 50 \quad 50)$$

given B is invertible the market is free of arbitrage and complete which means we can simply replicate the cashflow and price accordingly we can simply recreate

$$B = \begin{pmatrix} 105 & 0 & 0 & 0 \\ 3 & 103 & 0 & 0 \\ 0 & 0 & 0 & 100 \\ 4 & 4 & 104 & 0 \end{pmatrix}$$

$$(50 \quad 50 \quad 50 \quad 50) \begin{pmatrix} 105 & 0 & 0 & 0 \\ 3 & 103 & 0 & 0 \\ 0 & 0 & 0 & 100 \\ 4 & 4 & 104 & 0 \end{pmatrix}^{-1} \approx (0,445 \quad 0,467 \quad 0,5 \quad 0,481)$$

$$P_{B_{ann}}^f = (0,445 \quad 0,467 \quad 0,5 \quad 0,481) \begin{pmatrix} 100,96 \\ 98,11 \\ 85,48 \\ 100 \end{pmatrix} \approx 181,6$$

- 7) The price of the bond is 183.25 is there an arbitrage opportunity?
since we just build a product in the market with the exact same cash flows we just need to look at the prices

$$P_{B_{ann}}^f = 181,6 < 183,25 = P_{B_{ann}}$$

Thus we can simply sell the bond and buy the replicating portfolio making a profit

$$183,25 - 181,6 \approx 1,65$$

Simply said yes there is an arbitrage opportunity since the law of one price doesn't hold

And we sell the bond and buy the cashflow in the market